AfyaPlus Generative Layer ? Engineering Decision Memorandum

To: CTO and Medical Director
From: Platform Engineering (observability capstone)
Re: Six weeks of production AI without automated evaluation, drift monitoring, or cost tracking
Evaluation run mode: offline

Executive summary

The AfyaPlus generative layer has been answering clinical and insurance queries without automated quality gates, drift reports, or token accounting. We evaluated both gpt-4o-mini and gpt-4o on a 15-question clinical set (30 model rows), simulated three months of traffic, and projected 30-day spend at 2,000 requests/day. Mean judge overall is 4.23/5 for mini and 4.81/5 for gpt-4o; Month 3 mean ROUGE-L is 0.3389 versus 0.4064 in Month 1; 30-day generative spend is $25.22 against a $1500 cap.

Quality performance breakdown

Automated overlap (BLEU / ROUGE-L / token F1) plus an LLM judge (correctness, groundedness, relevance, helpfulness) were scored for every row. gpt-4o-mini passed 2 of 3 feature gates; gpt-4o passed 3 of 3. Mean groundedness is 4.47 (mini) vs 5.00 (gpt-4o). Clinically this means mini is cheaper on overlap-heavy protocol restatements but is more likely to drop numbers, waiting periods, or red-flag lists ? the details clinicians need for safe routing.

Cost and efficiency analysis

At the documented 75/25 canary, 30-day token spend is $25.22 ($0.84/day vs a $50 daily cap). Routing each feature to the cheapest model that still PASSES the quality gate saves $58.96 versus sending all traffic to gpt-4o and $-5.91 versus the current canary. Human review of the same 59,940 tickets at an assumed $2.50/ticket would cost $149,850 ? token inference remains far cheaper than full human validation, but it does not replace a clinician on failed gates.

Recommended routing from this run: chronic_care_referral ? gpt-4o; insurance_routing ? gpt-4o-mini; pediatric_triage ? gpt-4o-mini.

Systemic operational risks

First detected drift: Month 1, column input_length. Month 3 latency mean is 1101.6 ms with ROUGE-L 0.3389. That pattern matches longer, more complex production questions than the pre-deploy baseline. Hallucination risk is highest where groundedness fails the 4.0 gate: unsupported dosages or cover terms must not ship. Waiting-period emergencies remain a policy exception (bypass cover checks) and must stay out of cost-only routing.

Actionable engineering roadmap

1. Route by feature, not by default canary. Keep any FAIL feature on gpt-4o. Estimated 30-day saving vs all-gpt-4o: $58.96, with groundedness held at the 4.0 safety bar.
2. Page on first-column drift. Alert already isolates Month 1 / input_length. Wire the Prometheus `/metrics` gauges to the existing 80%/95% budget WARN/CRIT policy so spend and quality share one on-call path.
3. Do not drop human review on pediatric_triage FAIL rows. Token cost of the full 30-day corpus is $25.22 versus $149,850 for blanket human review; spend the human budget only on gate failures and Month 3 drifted slices.

Coursework snapshot ? not a live AfyaPlus production certification.